

Lab 4 – VibeCoding

Traffic Escape — Chat History

Samhitha Alluru | PES1UG24CS050

Today 11:40 AM

I have to do Lab 4 VibeCoding and I have been assigned the Traffic Escape repository. The README has the task requirements and the deliverables. I want to do the work properly from the beginning, so guide me step by step and go to the next step only after I have finished the previous one.

Yes. We can do this step by step. We will first get the repository running exactly as provided, understand the existing game, and record the original version before making changes. Then we will implement each requested feature one at a time, test it, and keep the changes organised with Git.

I have the repository open in VS Code. What should I check first?

First, check the README and project structure. Then make sure the terminal is opened in the project directory. Before changing any code, we need to run the original game successfully.

The project is in my Downloads folder and I opened the inner 50-traffic-escape folder in VS Code. I also need to make sure the original version stays unchanged because I need the before video.

That's correct. Keep the original copy separate and use your working repository for the feature changes. Run the original game first and record the before version before modifying the code.

I had an issue installing pygame because the project has pygame 2.5.2 and my Python version is 3.14.3. What should I do without changing the actual game behaviour?

The old pygame release is not building cleanly with that Python version. Use pygame-ce as the compatible Pygame implementation for your environment. This is a dependency/setup change so the existing game can run; it does not add one of the lab features.

The game is running now. I recorded the original version. What next?

Good. Now we can start the Vibe Coding part. Do one feature at a time and test after every change. For each prompt, clearly tell the coding assistant what to implement, what existing behaviour to preserve, and which later tasks it must not implement yet.

Task 1 — Three-life system

okay task 1 is the 3 lives feature. give me a proper prompt that i can directly use for vibe coding. make it specific and make sure it only does task 1 and doesn't change anything else.

Use this prompt:

```
Can you add a 3-life system to the existing Traffic Escape game? The player should start with exactly 3 lives and the remaining lives should be clearly displayed on the screen. Whenever the player collides with traffic, one life should be deducted and the player should be reset appropriately. The game should continue while at least one life remains and should only enter the existing game-over state when all 3 lives are lost. Please preserve the current player controls, traffic behaviour, scoring and overall gameplay. Do not implement the moving raft, persistent high-score system or day/night cycle yet, and do not make unrelated changes. Keep the change limited to what is necessary for this feature.
```

okay i applied it. how should i test it before committing?

Run the game and check that the player starts with 3 lives. Cause one collision and verify that the count decreases by one. Repeat the test and make sure the game does not end while lives are still available. Finally verify that the existing controls, traffic and scoring still work.

it works. i also see `__pycache__` changed. should i commit that?

No. Do not commit generated Python cache files. Stage only the source file that was changed for this feature.

okay what commands should i use?

Use:

```
git add game/game_engine.py
git commit -m "Add three-life system"
git push origin master
```

done. task 1 is pushed. next is the moving raft.

Task 2 — Moving raft lane

task 2 needs a moving raft/log lane. give me the prompt in the same way, but make sure the 3 lives feature stays unchanged.

Use this prompt:

```
Can you add a moving raft/log lane to the existing Traffic Escape game? The raft should continuously move across the safe lane, and the player should be able to get onto it and be carried along with its movement. Please preserve the 3-life system already implemented in Task 1, including the life display and collision behaviour. Also preserve the existing player controls, traffic behaviour, scoring and overall gameplay. Do not implement the persistent high-score or day/night features yet, and do not make unrelated changes. Keep the implementation limited to the moving raft functionality.
```

okay it is working. what should i test exactly?

Check that the raft moves across the lane, the player can stand on it, and the player is carried with it. Then verify that the three-life system still works after the raft feature was added.

done. how do i commit only task 2?

Stage the game engine file and create a separate commit:

```
git add game/game_engine.py
git commit -m "Add moving raft lane"
git push origin master
```

Task 3 — Persistent top-five high scores

task 3 is top 5 high scores. i need the scores to stay even after i close and reopen the game. give me a proper vibe coding prompt and don't change the previous features.

Use this prompt:

```
Can you add a persistent top-five high-score system to the game? Keep track of the player's score and maintain only the five highest scores, sorted from highest to lowest. Display the top five scores in the game and save them to a JSON file so they remain available after the game is closed and opened again. Load the existing scores when the game starts, handle the case where the JSON file does not exist yet, and update the file when a qualifying score is achieved. Please preserve the three-life system, moving raft, traffic, controls and existing scoring behaviour. Do not implement the day/night feature yet and do not make unrelated changes.
```

it created high_scores.json. is that supposed to be included?

Yes. The JSON file is the persistent storage for the top-five scores, so it is part of the updated project. Test that scores survive restarting the game and that only the best five scores are retained.

okay it works. what should i stage for this one?

Stage both the modified game engine and the new JSON file. Do not include `__pycache__` or other generated files.

commands?

```
git add game/game_engine.py high_scores.json
git commit -m "Add persistent top five high scores"
git push origin master
```

done. next is task 4.

Task 4 — Day/night cycle

task 4 is the day night cycle. it should automatically change every 30 seconds. give me the prompt and make sure all 3 previous features stay working.

Use this prompt:

```
Can you add an automatic day/night cycle to the existing Traffic Escape game? The game should switch between day and night every 30 seconds and the visual appearance should clearly change between the two modes. During night mode, make the headlights visible farther ahead so the player has better visibility. Please preserve all previously implemented features, including the three-life system, moving raft lane and persistent top-five high scores. Keep the existing player controls, traffic behaviour, scoring and gameplay unchanged. Do not introduce unrelated changes or remove any existing functionality. Only make the changes needed for the day/night feature.
```

task 4 is done. what should i check before i make the final video?

Run the complete game and check all four features together. Confirm the three-life system, moving raft, persistent high scores, and automatic day/night cycle all work. Also make sure the original controls and traffic behaviour still work.

everything is working. now i need to commit task 4 and push it.

Stage only the intended source changes, then commit and push the final feature:

```
git add game/game_engine.py
git commit -m "Add day-night cycle"
git push origin master
```

Final testing and submission preparation

all 4 tasks are done and pushed. i want to make sure i don't accidentally submit the original code as the updated code. how do i check that i am on the final version?

First check that you are on the master branch and that the latest commits are present. Your history should contain the initial setup commit followed by the four feature commits. Then run the final project one more time before making the after video.

what should the final code zip contain?

Use the final working project folder. Include the Python source files, README and required project data such as high_scores.json. Do not include .git, __pycache__ or .pyc files in the submission ZIP.

okay i have the final code backed up. now i need the after video and chat history.

For the after video, run the completed version and record about 10 seconds showing the final game. For the chat history, keep this conversation with the Vibe Coding prompts and the iterative testing process. Your final submission materials should include the before video, after video, updated code and the chat history required by the lab.

I also want to keep the prompts in the chat history exactly because that is the important part of showing the Vibe Coding process.

Yes. The prompts are an important part of the development record because they show how each feature was requested and how the changes were constrained. Keep the feature prompts together with the surrounding testing and follow-up messages rather than submitting only the prompts by themselves.

End of chat history